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Ethnomedicinal plants used to treat epilepsy in Jharkhand state

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Abstract : Epilepsy is a neuropsychological disorder, caused due to discharge of neurotransmitter. From time immemorial, plants have been used for their healing and therapeutic abilities, thus healing action. The aim of the present study was to highlight the anti epileptic activity of some medicinally important plants of Jharkhand state, like *Acorus calamus, Artemisia vulgaris Bacopa monnerei, Centella asiatica, Caesalpinia cristaand* and many more. The role of such plants, with specific properties of their parts have been documented through exploration and person dialogue. This paper reviews the potential of such plants that can be explored to ascertain anti epileptic activity.

Key Words: Epilepsy, neuropsychological, neurotransmitter, therapeutic, *Acorus calamus*, *Centella asiatica*

INTRODUCTION

Herbal medicine is a complementary or alternative system of medicinal practiced all over the world since the beginning of human civilization. The practice of traditional system of medicines gradually waned with the introduction of modern system of allopath. Today herbal remedies are back into prominence. Till now, natural product compounds discovered from medicinal plants (and their analogues also) have provided numerous clinically useful drugs. Four billion people or about 80% of the world's population uses herbal medicine as part of health care (WHO, 2011)[1].

According to the diagnostic and Statistical Manual of Mental Disorders-Fourth Edition, a mental disorder is any clinically significant behavioural or psychological syndrome characterised by the presence of distressing symptoms, impairments of functioning, a significantly increased risk of suffering death, pain, disability or loss of freedom.

Epilepsy is one of the very common mental disorders that affects about 50 million people worldwide.[2] Nearly 80% of the people with epilepsy are found in developing regions (WHO, 2002)[3].

Epilepsy is a condition, which causes seizures/ convulsion to occur. It is one of the most common chronic diseases affecting human beings. It is a neuropsychological disorder which occurs due to over discharge of neurotransmitter substance.

The state of Jharkhand is inhabited by large population of ethnic communities, living in true harmony with nature and entirely dependent upon plants and plant products for their livelihood and primary health care. Various developmental programs by the government to create modern amenities or other human related factors have direct or indirect onslaught on the forests and forest dwellers. Forest is completely disappearing and people living in the abode of forests or remote and primitive villages are forced to leave their homelands where they were living for thousands of years with their rich cultural heritage. In fact, these people being very close to nature are true repository of knowledge, who are also compelled to migrate another area and to adopt alien profession forgetting inherited knowledge which they acquired from their predecessors from words of mouth and preserved for generations together with practice only.

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The Jharkhand being a nascent state various developmental schemes are underway to give it a new face. In the state the education is very poor, majority of the population is under poverty line, socio-economic condition is very poor and living condition is unhygienic. Due to lack of proper medical facilities at close door or some other reasons majority of the people rely on traditional system of medicine/ Horopathy (tribal system of medicine). People with epilepsy and their families suffer from stigma and discrimination in many parts of the world. It can affect people economically, socially and culturally Specially, in the tribal communities patients with epilepsy have been stigmatized, shunned, or even imprisoned. However, there are large number of medications in allopathy, but due to high cost and many side effect one looks for an alternative medicine. In this regard traditional system with herbal drugs gives great relief because of low cost and little side effect.

In the changed scenario of unscrupulous and unplanned activities our valuable plant are under various degree of threats and also the tribes and indigenous people, who are our knowledge resource are on run.

Therefore, the present investigation has been undertaken with objectives to interact with local traditional healers, old knowledgeable persons and document their knowledge on medicinal plants, their usage and the types of mental diseases treated in general and epilectic disorder in particular. So far no systematic ethnobotanical survey has been made in this area and this is the first report on the medicinal plants used by the local traditional healers.

STUDY AREA

The Jharkhand State is the homeland of the tribes and is very rich in terms of ethnic diversity and natural resources. It extends between 22°00' to 24°37' N latitude and 83°15' to 87°01' E longitude. Forest occupies more than 29% of the state which is amongst the highest in India. The altitude of the Jharkhand plateau ranges between 1000 -3000 ft above the mean sea level. Jharkhand is surrounded by five states, namely West Bengal, Chhattisgarh, Odisha, Uttar Pradesh and Bihar.

The Jharkhand state has total surface area of 79,714 Sq km including 24 districts, 212 blocks and 32,620 villages with a population density of 338 people per Sq km. It is inhabited by 32 tribal communities viz., Oraon, Munda, Mahli, Baraik, Bhumij, Goriat, Karmali, Bedia, Bhuiya, Lohra, Santhal, Ho, Gond, Kharia, Korba and various indigenous caste such as Mahto, Gonjhu, Ghansi, Teli,etc. According to 2001 census Jharkhand is having total population of 26,90,9428, comprising of 26.3% Schedule Tribes, 12% Schedule Caste, 27.9% Backward Classes and rest others.

MATERIALS AND METHODS

The investigation was carried out continuously from 2008 - 2010. The plant materials of ethno-medicinal importance were collected from different sites of study areas and identified with their botanical names based on local names. Several books on Indigenous Medicinal Plants were consulted for correct information and verification. However, renowned taxonomist , were also contacted for identification of plants. Herbaria were prepared adopting standard practice and preserved in the University Department of Botany for reference. Plants were photographed in situ and herbaria were also photographed.

For the survey work 25 villages were selected prioritizing the ethnic population especially the tribal groups. Surveys to villages were done at regular interval for collecting plant specimens, related therapeutic information and to have photographs of *in situ* plants. Local persons were instrumental to overcome the language problems and to win the confidence of informants (patients, vaidhyas, pahans and kabiraj) as they generally deny to share their ideas with the outsiders. Most of the remote villages were surveyed three times while those in the more accessible areas were visited several times with an aim to obtain authentic information and to collect plant specimens in their appropriate growing seasons. Tools used for routine visit to different place of significance are; detail map of the study area, questionnaire, digital camera, canvas bags and magnifying glasses.

RESULTS AND DISCUSSION

In the present investigation 25 medicinal plants have been described in table 1 with their parts used for epileptic disorders, like covulsion, seizzure. Different parts, such as root, rhizome, stem, leaves, flowers, fruits, seeds, bark or the whole plant is used to treat the epilepsy. Usually, for drug preparation different parts of the same plant are mixed together or two or parts of two or more plants

Plant species belong to 19 families and may be herbs, shrubs or trees. The plant drug is given in different forms like powder, decoction, infusion, paste or pills. Rhizome of *Acorus calamus* and *Curcuma zedoaria* are prescribed for mental disorders in horopathy, specially the use of former is more common. Whole plant is used for preparation of herbal drugs for e.g., *Amaranthus virdis*, *Bacopa monnieri*, *Centella asiatic* , *Cynodon dactylon*, *Datural metel*, *Moringa oleifera*,*Oxalis corniculata*,*Piper longum* , *Sida cordifolia* , *Solanum nigrum.* All the plants enlisted are also reported to be beneficial in other types of mental disorders, such as anxiety, hysteria, insomnia, depression, headache, memory enhancer and dizziness etc.

Ethnobotany is not new to India because of its rich ethnic diversity. Jain (1991)[4] pointed out that there are over 400 different tribal and other ethnic groups in India. The tribes constitute about 7.5 percent of India's population. During the last few decades there has been an increasing interest in the study of medicinal plants and their traditional use in different parts of India and there are many reports on the use of plants in traditional healing by either tribal people or indigenous communities of India.[5-9] However, reports are also available on plants used in mental disorders.[10-14]

During the interviews with traditional medicine practitioners as part of the fieldwork to collect ethnobotanical data, it was possible to identify two main different types of practices in tribal society: one resorts to aspects of spirituality, evil spirit, witchcraft or exorcism and the second takes into account the knowledge of the medicinal properties of plants and other therapeutic techniques, while the third one combines the use of medicinal plants with rituals and prayers.

A perusal of the literature reveals that ethnomedicinal works or Traditional medicinal practices have remained as a component of health care system of many countries all over the world[15,16,17,18,19,20,21,22] because these measures fit into the cultures of people and are not usually as expensive and do not possess many side effects, contraindications and possible interactions with drugs used simultaneously.

The present investigation together with other related information are the first level of validation of medicinal properties of plants, however, biochemical investigation, toxicology, and preclinical and clinical trials are preconditions before final verdict to be used for mankind.

CONCLUSION

The survey indicated that, the study area has plenty of medicinal plants to treat a wide spectrum of human ailments. Earlier studies on traditional medicinal plants also revealed that the economically backward local and tribal people of Jharkhand state widely use folk medicine due to low cost and sometimes it is a part of their social life and culture. It is evident from the interviews conducted in different villages, knowledge of medicinal plants is limited to traditional healers, herbalists and elderly persons who are living in rural areas. This study also points out that certain species of medicinal plants are being exploited by the local residents who are unaware of the importance of medicinal plants in the ecosystem.

Nevertheless the majority of the cases remain undetected and untreated, both by the health services and traditional medicine. On the other hand the neurologic disorders are looked at by the public as a result of occult forces or witchcraft, which does not help a correct diagnosis and correct treatment for these types of problems.

It is therefore important to promote clarity of the general local public, particularly the practitioners of traditional medicine, on the causes, symptoms, and the possibilities of treatments for the neurologic disorders. This would help to demystify the neurologic and mental disorders, decrease the isolation and exclusion of the patients suffering these types of disorders and promote an adequate treatment whenever possible.

Taking into account the existing knowledge in the country on the medicinal properties of plants, it is believed that research in the area of ethnomedicine and ethnopharmacology in the Jharkhand state must proceed in order to bring about promising results that can add value to the very rich natural resources of this state.

Table 1: Plants used in epilepsy with their botanical names, families, local names and parts used

Sl. No.	Botanical Name	Family	Local Name	Part Used	Disease
1	*Abrus precatorius* Linn.	Fabaceae	Gunja	Root, Leaves, Bark, Seeds	Convulsion
2	*Acorus calamus* Linn.	Araceae	Bach	Rhizome	Epilepsy, Hysteria
3	*Amaranthus virdis* Linn.	Amaranthaceae	Wild chaulai	Whole plant	Convulsion, Epilepsy
4	*Artemisia vulgaris* Linn	Asteraceae	Dauna	Root, leaves, Flowering top	Hysteria, epilepsy
5	*Bacopa monnieri* Linn. Wettst	Scrophulariaceae	Jalneem	Whole plant	Insanity, Epilepsy, mental weakness
6	*Caesalpinia crista* Linn.	Caesalpinaceae	Latakaranja	Bark, Seed, Leaves	Convulsion, Epilepsy
7	*Carrisa carandus* Linn.	Apocynaceae	Karaunda	Root, Leaves, Fruit	Epilepsy
8	*Cassia occidentalis* Linn.	Fabaceae	Kashmard	Leaf, Flower, Seed	Epilepsy, Hysteria
9	*Celestrus paniculatus* Willd.	Celestraceae	Malkagni	Bark root, Leaves, Seed	Brain Tonic, Headache
10	*Centella asiatica* (Linn)	Apiaceae	Manduk parni	Whole Plant	Epilepsy, Hysteria, Enhance memory power
11	*Clitoria ternatea* Linn.	Fabaceae	Aparajita	Root, Seed	Syncope, Vertigo, Brain weakness, epilepsy
12	*Curcuma zedoaria* Linn.	Zingiberaceae	Chandragul	Rhizome	Dizziness, Epilepsy
13	*Cynodon dactylon* Linn. Pers.	Poaceae	Doob grass	Whole plant	Hysteria, Epilepsy
14	*Datural metel* Linn.	Solanaceae	Datura	Whole Plant	Epilepsy, Hysteria, Migraine
15	*Glycyrrhiza glabra* Linn.	Fabaceae	Mulethee	Dried stem or Root	Epilepsy
16	*Moringa oleifera* Lamk	Moringaceae	Sahjan, Munga	Whole plant	Epilepsy, nervous debility
17	*Oxalis corniculata* Linn.	Oxilidaceae	Amita Sag	Whole plant	Convulsion
18	*Papaver somniferum* Linn.	Papaveraceae	Afim	Seeds, Flower	Epilepsy, Hysteria, Insomnia
19	*Piper Longum* Linn.	Piperaceae	Pipalli	Whole plant	Insomnia, epilepsy
20	*Psidium guajava* Linn.	Myrtaceae	Amrud	Leaves	Convulsion, Epilepsy
21	*Ricinus communis* Linn.	Euphorbiaceae	Arand	Leaves, Flower, Seed	Epilepsy, convulsion
22	*Semecarpus anacardium* Linn.	Anacardiaceae	Bhelwa	Seed, Fruit	Anxiety, Epilepsy
23	*Sida cordifolia* Linn	Malvaceae	Bariyar	Whole plant	Epilepsy
24	*Solanum nigrum* Linn	Solanaceae	Makoy	Whole plant	Convulsion, Insomnia
25	*Strychous nux-vomica* Linn.	Loganiaceae	Kuchla	Dried riped Seed, Bark	Anxiety, Epilepsy Hysteria

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